

This thesis uses a systematic literature review to address its research question. The decision comes from the question itself:

whether and under what conditions firm-level AI investments are profitable is an empirical issue on which data has rapidly accumulated across various outlets. This method is mostly chosen by researchers because of its comprehensiveness and transparency (Rojon et al., 2021, p. 196)¹. To allow for the reconstruction of each decision, every step that shapes the final corpus is recorded here. The only data source is Scopus, which was accessed through the WU license. One recorded query is the source of every study that is included in the review, therefore all of the review's conclusions are based entirely on the corpus.

Other sources are only mentioned as methodology literature in this chapter and as theory anchors in Chapter 2, this review is restricted to peer-reviewed journal publications, no preprints or gray literature. The search query was run on July 17 2026 through the Scopus Search API with the following Query: TITLE(("artificial intelligence" OR "machine learning" OR "deep learning" OR "AI")) AND TITLE-ABS-KEY(("invest" OR "invests" OR "invested" OR "investing" OR "investment" OR "investments" OR "investor" OR "investors" OR adopt* OR implement*) AND ("competitive advantage" OR "firm performance" OR "financial performance" OR "firm productivity" OR "firm value" OR "market value" OR "firm growth")) AND SUBJAREA(BUSI OR ECON OR DECI) AND PUBYEAR > 2014 AND PUBYEAR < 2027 AND DOCTYPE(ar) AND LANGUAGE(english) AND SRCTYPE(j)

The requirement that the AI terms have to appear in the title limits the search to studies where AI is the primary focus rather than a side topic. The investment and outcome vocabulary is searched more broadly, in title, abstract and keywords, because outcome measures are often named only in the abstract. Only terms are matched by Scopus, not concepts; a search phrase is not expanded to its synonyms, so a query looking simply for firm performance might overlook studies that refer to the same outcome as firm growth or firm value. Because a number of operations research publications publish firm-level AI evidence and are indexed there rather than under business or economics, decision sciences is included as a subject area. Additionally instead of only using the wildcard term "invest", the investment terms are expressed word for word. An earlier version of the query found 75 additional records that only looked into AI without considering it as an investment, because the "invest" term also matches investigate. A sound query must find Babina et al.

(2024), Krakowski et al. (2023), and Lui et al. (2022), three studies that were set as retrieval benchmarks. The final query returns 432 different results including all three benchmark studies.

What stays in the corpus is determined by two criteria.

The first is journal quality: a record is kept if its journal has a rating of at least 2 according to the Academic Journal Guide (AJG). The majority of the records that this rule eliminates came from journals that aren't even listed in the ranking at all. Every record, regardless of the year of publication, is subject to a single, fixed edition of the ranking, which maintains consistency and non-selectivity. The single source of truth for this thesis is the AJG 2024 edition. The second layer has 4 criteria. A study has to be (1) empirical, (2) at the firm level, (3) focused on the firms AI investment or adoption as its object of study, and (4) measuring a performance or competitive-advantage outcome.

Only when all are met can the record stay in. Case studies are regarded as empirical and studies that survey managers stay in when the outcome is measured at the firm level.

In order to ensure that each excluded record has a single reason so the counts are clear, exclusions are assigned one of five codes in a fixed order with the first matching code gets assigned: E1 not empirical (conceptual work, reviews, simulations without firm data); E2 not at the firm level (individual, team, or country); E3 AI is the study's analysis tool rather than the firms investment object; E4 no performance or competitive-advantage outcome (for example adoption intention or user satisfaction); E5 wrong publication type.

Across all criteria, a record that left doubt at the title and abstract stage was kept in for further review.

Figure 3.1 shows the entire process. First the 432 records were reduced to 174 using the journal-quality filter, the record did not meet the minimum AJG ranking requirements. 93 of the 174 remaining records (E1: 20, E2: 7, E3: 39, E4: 27; E5 was never needed) were eliminated after the abstract and title were compared against the criteria. With the remaining 81 records the texts were then evaluated for eligibility using the same criteria, which were determined by an article's method and results sections, another 14 more were

excluded in this stage: 12 on the basis of unfitting content, and two whose texts were not available via open access or the university license. In total 67 studies are included in the review.

As with the coding described below, screening was AI-assisted, with the authors review and confirmation on every decision.

Each included study occupies one row of the coding table, split into study characteristics and findings. The scheme was fixed before coding began, with the fields adapted to the research question instead of adhering to a single template. The characteristics side documents how the evidence was produced, including its method, its sample and time frame, the country and industry it covers, the theoretical lens a study uses and how it measures AI investment, such as through surveys, AI-related job postings or adoption announcements.

The findings side records what the evidence says. It holds the outcome: performance, competitive advantage, or both. Performance means the firm improves by its own numbers, such as sales, profit, or productivity; competitive advantage means the firm does better than its competitors. Each construct has its own measurement field. The findings side records the direction of the effect, the identified conditions with their roles, a one-sentence key finding, and quality notes, for example when an outcome rests on managers self-reports.

The core fields of this table are printed in the appendix (Tables A.1 and A.2). The theoretical lens, industry, and quality notes are incorporated into the thesis through the descriptive mapping and the discussion.

There are five possible options for the effect's direction. The average effect, can be described as positive, negative, or null. Mixed means the finding splits across outcome measures, as when revenue rises but costs do not fall. Conditional means that the effect's size or sign relies on discovered conditions. Because they address different questions, direction indicates whether the investment pays on average, while conditions indicates what the payoff depends on.

Direction and conditions are coded in different columns, as a result, a study might be classified as positive while still listing conditions. The conditions just enhance or reduce the effect on average.

Two general rules: A sample period or industry that is never mentioned in the publication remains null rather than being inferred since a cell only includes what a study reports. When a study's value falls outside of any of the categories, the cell contains the study's original text.

Each study is coded according to whether it measures competitive advantage, superior business performance, or both. Only when a study analyzes competitive advantage through a validated competitive-advantage scale, or using data comparing the company to its rivals does it qualify as measuring competitive advantage. Performance is coded as a performance gain alone. The first example is demonstrated by Chatterjee et al. (2021), who define competitive advantage in relation to other businesses in the same market. The second is demonstrated by Babina et al. (2024): companies who invest in AI see faster growth in sales and employment. The study does not claim a position in relation to competitors, thus it is coded as performance. The table shows how rare it is for competitive advantages to be measured. Every study was coded twice in independent, AI-assisted runs: Claude Fable 5 (Anthropic) drafted all rows as the first coder, and Gemini 2.5 Pro (Google) coded the same studies as the second, blind to the first run.

Agreement between the runs was 90% for method, 88% for the outcome construct, and 73% for effect direction. The author checked every study's coded values against the results reported in the paper, including the studies on which the runs agreed, to confirm that the coding was justified. For the 29 studies on which the runs disagreed on at least one coded field, the author read the text and reached an independent decision before looking at either run's values, then resolved the disagreement against decision rules recorded in writing.

The conditions column also provides the structure of the synthesis in Section 4.2: every condition entry was assigned to one of six thematic groups, and a study stands in several groups when it identifies several conditions. The grouping followed the same dual principle as the coding.